

1. (a) The first moment about origin and second moment about mean of a characteristic of 100 items were found to be 60 and 100 respectively. At calculation stage two items were wrongly taken as 5 and 4.5 instead of 30 and 20. Calculate the correct coefficient of variation of the characteristic. (4 marks)

(b) For a distribution, the mean is 10, variance is 16, γ_1 is +1 and β_2 is 4. Obtain the first four moments about origin. Comment on the nature of skewness and kurtosis. (4 marks)

1(a) $n = 100$
 $\mu_1' = \bar{x} = 60, \mu_2 = \sigma^2 = 100$

Wrong observations 5, 4.5

Correct observations 30, 20

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i' \Rightarrow n\bar{x} = \sum_{i=1}^n x_i'$$

Hence wrong sum $\sum_{i=1}^n x_i' = 100 \times 60 = 6000$

Correct sum $\sum_{i=1}^n x_i' = 6000 - 5 - 4.5 + 30 + 20 = 6040.5$

Hence correct Mean $\bar{x}_c = \frac{6040.5}{100} = 60.405 \rightarrow (1 \text{ mark})$

Similarly $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i' - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^n x_i'^2 - \bar{x}^2$

$$\Rightarrow \sum_{i=1}^n x_i'^2 = n(\sigma^2 + \bar{x}^2)$$

Hence wrong $\sum_{i=1}^n x_i'^2 = 100(100 + 3600) = 370000$

Hence correct $\sum_{i=1}^n x_i'^2 = 370000 - (5)^2 - (4.5)^2 + (30)^2 + (20)^2$

$$= 370000 - 25 - 20.25 + 900 + 400$$

$$= 371254.75$$

Therefore correct $\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_{ic}^2 - \bar{x}_c^2$

$$= \frac{1}{100} (371254.75) - (60.405)^2$$

$$= 3712.5475 - 3648.764025$$

$$= 63.783475$$

Correct $\sigma = 7.986455722$

C.V = $\frac{\sigma}{\bar{x}} = \frac{7.9865}{60.405} = 0.132216 \rightarrow (1 \text{ mark})$

$= 7.9865 \rightarrow (2 \text{ marks})$

1(b)

$$\bar{x} = 10, \sigma^2 = 16, \gamma_1 = +1, \beta_2 = 4$$

Obtain first four moments about origin. Comment on the nature of skewness and kurtosis.

$$\mu_1' = \bar{x} = 10 \longrightarrow \underline{\underline{\frac{1}{2} \text{ mark}}}$$

$$\sigma^2 = \mu_2 = \mu_2' - \mu_1'^2 \Rightarrow \mu_2' = \mu_2 + \mu_1'^2 = 16 + 100 = 116$$

$\longrightarrow \underline{\underline{\frac{1}{2} \text{ mark}}}$

$$\gamma_1 = +1, \text{ we know } \gamma_1 = +\sqrt{\beta_1}$$

$$\Rightarrow \beta_1 = \gamma_1^2 = 1$$

$$\Rightarrow \frac{\mu_3^2}{\mu_2^3} = 1 \Rightarrow \mu_3^2 = \mu_2^3 = (16)^3 = 4096$$

$$\Rightarrow \mu_3 = 64$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\Rightarrow \mu_3' = \mu_3 + 3\mu_2'\mu_1' - 2\mu_1'^3$$

$$= 64 + 3 \times 116 \times 10 - 2 \times (10)^3$$

$$= 64 + 3480 - 2000 = 1544 \longrightarrow \underline{\underline{(1 \text{ mark})}}$$

$$\text{Given } \beta_2 = 4$$

$$\Rightarrow \frac{\mu_4}{\mu_2^2} = 4 \Rightarrow \mu_4 = 4\mu_2^2 = 4 \times (16)^2 = 1024$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$\Rightarrow \mu_4' = \mu_4 + 4\mu_3'\mu_1' - 6\mu_2'\mu_1'^2 + 3\mu_1'^4$$

$$= 1024 + 4 \times 1544 \times 10 - 6 \times 116 \times (10)^2 + 3 \times (10)^4$$

$$= 1024 + 61760 - 69600 + 30000$$

$$= 23184 \longrightarrow \underline{\underline{(1 \text{ mark})}}$$

$$\text{Hence, } \mu_1' = 10, \mu_2' = 116, \mu_3' = 1544, \mu_4' = 23184$$

$$\beta_1 = 1 \text{ and } \mu_3 > 0 \text{ +vely skewed} \longrightarrow \underline{\underline{\frac{1}{2} \text{ mark}}}$$

$$\beta_2 = 4 > 3 \text{ Leptokurtic (highly peaked)} \longrightarrow \underline{\underline{\frac{1}{2} \text{ mark}}}$$

2/a:
$$f(x) = \begin{cases} \frac{x^2}{81} & ; -3 < x < 6 \\ 0 & ; \text{otherwise} \end{cases}$$

$$U = \frac{1}{3}(x-12) \quad \text{for } x = -3, \quad U = -5$$

$$x = 6, \quad U = -2$$

$$x = 3U + 12$$

$$dx = 3du$$

$$f(u) = \begin{cases} \frac{3}{81} (3U+12)^2 & -5 < U < -2 \\ 0 & ; \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{3} (U+4)^2 & ; -5 < U < -2 \\ 0 & ; \text{otherwise} \end{cases}$$

Total (4)

2/b:

$$f(x,y) = \begin{cases} cxy & ; 0 < x < 4, \quad 1 < y < 5 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\iint f(x,y) dx dy = 1 = c \int_0^4 \int_1^5 xy dx dy = c \int_0^4 x \left(\frac{y^2}{2} \right)_1^5 dx = 12c \left(\frac{x^2}{2} \right)_0^4$$

$$1 = 96c$$

$$c = \frac{1}{96}$$

$$P(x+y \leq 3) = \int_1^3 \int_0^{3-y} \frac{1}{96} xy dx dy$$

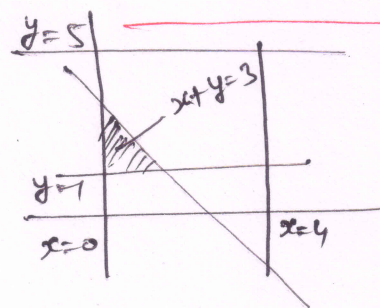
$$= \frac{1}{96} \int_1^3 y \left(\frac{x^2}{2} \right)_0^{3-y} dy$$

$$= \frac{1}{96} \int_1^3 y \left(\frac{9+y^2-6y}{2} \right) dy$$

$$= \frac{1}{2 \times 96} \left(\frac{9y^2}{2} + \frac{y^4}{4} - \frac{6y^3}{3} \right)_1^3$$

$$= \frac{1}{2 \times 96} (36 + 20 - 52)$$

$$= \frac{1}{48}$$



Total (4)

1. (a) The spectrum of the two-dimensional random variable (X, Y) is,
 $(x_i, y_j) = (i, j), (i = 0, 1, 2; j = 0, 1, 2)$ excluding $(0, 2)$ and $(2, 0)$.

The probability masses at different spectrum points along with marginal probability masses of X and Y are given by the following table :

X \ Y	0	1	2	P(Y=j)
0	$\frac{1}{8}$	$\frac{1}{8}$	0	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$
2	0	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
P(X=i)	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	1

The above table is obtained as follows:

The event space contains total 8 outcomes which are $(M, M, M), (M, M, F), (M, F, M), (F, M, M), (M, F, F), (F, M, F), (F, F, M), (F, F, F)$, where, F represents a 'Female birth' and M represents a 'Male birth'.

Therefore, the event $(X=1, Y=1) = \{(M, F, M), (F, M, F)\}$. And hence, $P(X=1, Y=1) = \frac{2}{8} = \frac{1}{4}$.

Similarly, the event, $(X=1, Y=0) = \{(F, M, M)\}$. And hence, $P(X=1, Y=0) = \frac{1}{8}$.

Again, the events $(X=0, Y=2)$ and $(X=2, Y=0)$ are impossible events and so on.

Now, the conditional distribution Y on the hypothesis of $X=1$ is given by p.m.f.,

$$p_{j|1} = \frac{p_{x=1, y=j}}{p_{x=1}} = 2p_{x=1, y=j} = 2p_{1j}$$

$$\text{Therefore, } P(Y=2 | X=1) = p_{2|1} = 2p_{12} = \frac{1}{4}$$

- (b) The general formula for the moment generating function of a discrete distribution is given by, $M_X(t) = \sum_x f(x)e^{tx}$, where the sum is over all values x of X . Again, the

expression given for moment generating function is $M(t) = \frac{1}{6}e^{-2t} + \frac{1}{3}e^{-t} + \frac{1}{4}e^t + \frac{1}{4}e^{2t}$.

Comparing these two expressions, we observe that X must take the values $-2, -1, 1$ and 2 with probabilities $\frac{1}{6}, \frac{1}{3}, \frac{1}{4}$ and $\frac{1}{4}$ respectively. Therefore,

$$P(|X| \leq 1) = P(X = -1) + P(X = 1) = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

Now,

$$M(t) = \frac{1}{6}e^{-2t} + \frac{1}{3}e^{-t} + \frac{1}{4}e^t + \frac{1}{4}e^{2t}$$

$$\Rightarrow M'(t) = -\frac{1}{3}e^{-2t} - \frac{1}{3}e^{-t} + \frac{1}{4}e^t + \frac{1}{2}e^{2t}$$

$$\Rightarrow M''(t) = \frac{2}{3}e^{-2t} + \frac{1}{3}e^{-t} + \frac{1}{4}e^t + e^{2t}$$

$$\Rightarrow M'''(t) = -\frac{4}{3}e^{-2t} - \frac{1}{3}e^{-t} + \frac{1}{4}e^t + 2e^{2t}$$

(1)

$$\text{Hence, } \mu_1 = -\frac{1}{3} - \frac{1}{3} + \frac{1}{4} + \frac{1}{2} = \frac{1}{12}$$

$$\mu_2 = \frac{2}{3} + \frac{1}{3} + \frac{1}{4} + 1 = \frac{9}{4}$$

$$\mu_3 = -\frac{4}{3} - \frac{1}{3} + \frac{1}{4} + 2 = \frac{7}{12}$$

(1)

Now, third central moment is,

$$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3 = \frac{7}{12} - 3 \times \frac{9}{4} \times \frac{1}{12} + 2 \times \left(\frac{1}{12}\right)^3 = \frac{19}{864}$$

(1)

$$1+1+1+1=4$$

Q 41. (a). pdf $f(x) = \frac{5}{x^6} \quad x \geq 1$

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_1^{\infty} x \frac{5}{x^6} dx = \frac{5}{4}$$

$$\begin{aligned} \sigma^2 = \text{Var}(X) &= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 \\ &= \int_1^{\infty} x^2 \cdot \frac{5}{x^6} dx - \left(\frac{5}{4}\right)^2 \\ &= \frac{5}{3} - \frac{25}{16} = \frac{5}{48} \end{aligned}$$

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$$\begin{aligned} P(X \geq 2.5) \\ \cancel{P(X)} &= P\left(X \geq \frac{5}{4} + \frac{5}{4}\right) \\ &= \cancel{P(X)} \end{aligned}$$

$$P\left(|X - \mu| \geq \frac{5}{4}\right) = P(X \geq 2.5 \text{ or } X \leq 0)$$

$$\begin{aligned} \Rightarrow P(X \geq 2.5) &= P\left(|X - \mu| \geq \frac{5}{4}\right) \leq \frac{\sigma^2}{\left(\frac{5}{4}\right)^2} = \frac{\frac{5/48}{25/16}}{1} \\ &= \frac{1}{15} \end{aligned}$$

① $P(X \geq a) \leq 0.15$

by Chebyshev's Inequality

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

$$\frac{15}{10} \quad 0.15 = \frac{\sigma^2}{\varepsilon^2} = \frac{5/48}{\varepsilon^2}$$

$$\Rightarrow \varepsilon = \sqrt{\frac{5/48}{0.15}} = \frac{5}{6}$$

$$P\left(\varepsilon - u < X \text{ or } X > u + \varepsilon\right)$$

$$P\left(|X - u| > \frac{5}{6}\right)$$

$$= P\left(-\frac{5}{6} + u < X \text{ or } X > u + \frac{5}{6}\right)$$

$$= P\left(-\frac{5}{6} + \frac{5}{4} < X \text{ or } X > \frac{5}{4} + \frac{5}{6}\right)$$

$$= P\left(\frac{5}{12} < X \text{ or } X > \frac{25}{12}\right)$$

$$= P\left(X > \frac{25}{12}\right) \text{ or } X > 1$$

$$\Rightarrow a = \frac{25}{12}$$

2

(b).

$$E(X_n) = 5$$

$$\text{Var}(X_n) = 2$$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(X_i) + \sum_{\substack{i=1, j=1 \\ i \neq j}}^n \text{Cov}(X_i, X_j) \right]$$

$$\text{Cov}(X_i, X_j) < 0 \quad \forall i, j \in \{1, \dots, n\}$$

$$\text{Var}(\bar{X}_n) \leq \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{2}{n}$$

$$P(|\bar{X}_n - 5| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} \leq \frac{2}{n\varepsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$P(|\bar{X}_n - 5| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$\Rightarrow$

$$\Rightarrow \bar{X}_n \xrightarrow{P} 5$$

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